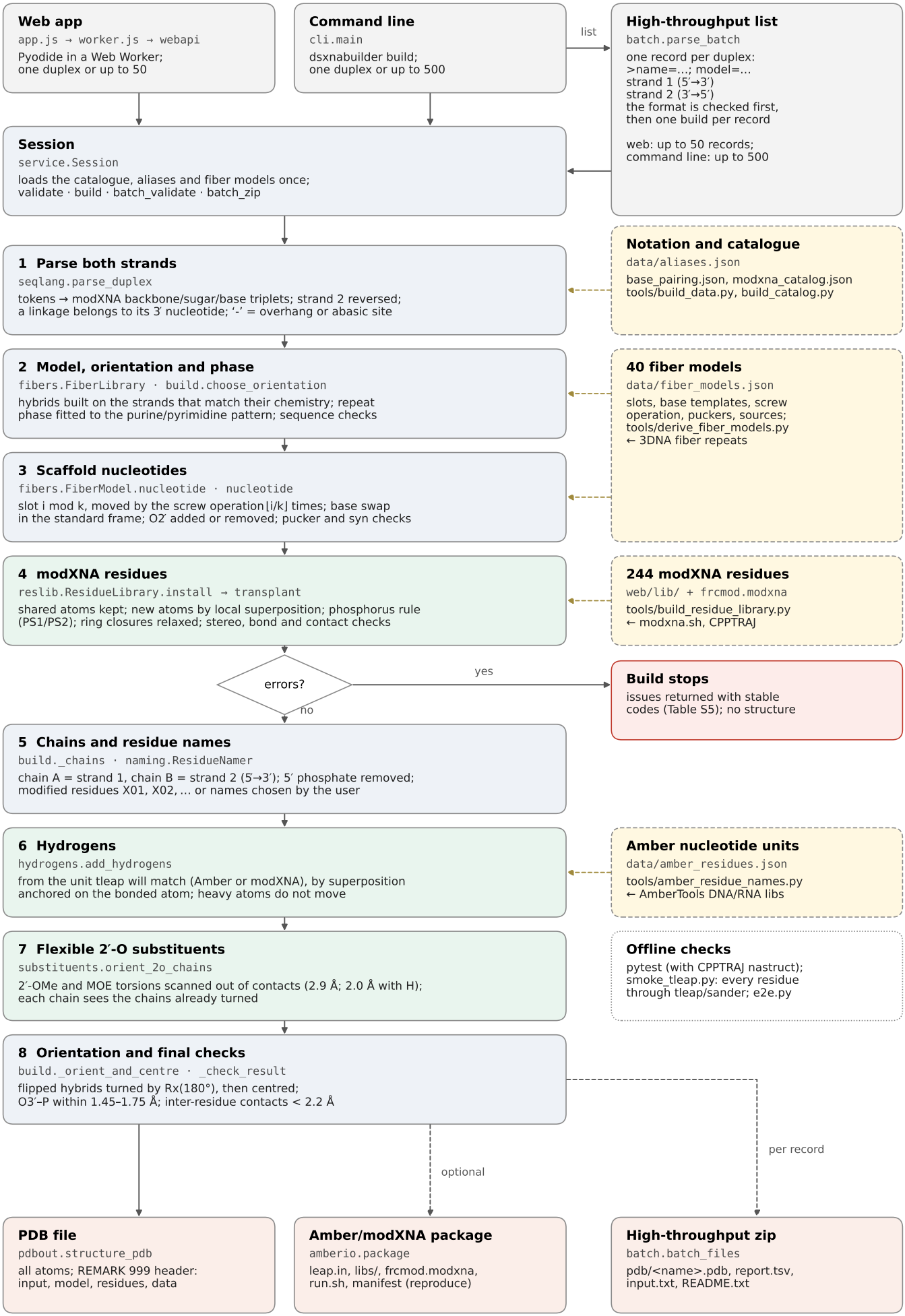

Web app
app.js → worker.js → webapi
Pyodide in a Web Worker;
one duplex or up to 50
Command line
cli.main
dsxnabuilder build;
one duplex or up to 500
list
High-throughput list
batch.parse_batch
one record per duplex:
>name=...; model=...
strand 1 (5′→3′)
strand 2 (3′→5′)
the format is checked first,
then one build per record
web: up to 50 records;
command line: up to 500
Session
service.Session
loads the catalogue, aliases and fiber models once;
validate · build · batch_validate · batch_zip
Notation and catalogue
data/aliases.json
base_pairing.json, modxna_catalog.json
tools/build_data.py, build_catalog.py
1 Parse both strands
seqlang.parse_duplex
tokens → modXNA backbone/sugar/base triplets; strand 2 reversed;
a linkage belongs to its 3′ nucleotide; '-' = overhang or abasic site
2 Model, orientation and phase
fibers.FiberLibrary · build.choose_orientation
hybrids built on the strands that match their chemistry; repeat
phase fitted to the purine/pyrimidine pattern; sequence checks
40 fiber models
data/fiber_models.json
slots, base templates, screw
operation, puckers, sources;
tools/derive_fiber_models.py
← 3DNA fiber repeats
3 Scaffold nucleotides
fibers.FiberModel.nucleotide · nucleotide
slot i mod k, moved by the screw operation [i/k] times; base swap
in the standard frame; O2′ added or removed; pucker and syn checks
4 modXNA residues
reslib.ResidueLibrary.install → transplant
shared atoms kept; new atoms by local superposition; phosphorus rule
(PS1/PS2); ring closures relaxed; stereo, bond and contact checks
244 modXNA residues
web/lib/ + frcmod.modxna
tools/build_residue_library.py
← modxna.sh, CPPTRAJ
errors?
yes
no
Build stops
issues returned with stable
codes (Table S5); no structure
5 Chains and residue names
build._chains · naming.ResidueNamer
chain A = strand 1, chain B = strand 2 (5′→3′); 5′ phosphate removed;
modified residues X01, X02, ... or names chosen by the user
6 Hydrogens
hydrogens.add_hydrogens
from the unit tleap will match (Amber or modXNA), by superposition
anchored on the bonded atom; heavy atoms do not move
Amber nucleotide units
data/amber_residues.json
tools/amber_residue_names.py
← AmberTools DNA/RNA libs
7 Flexible 2′-O substituents
substituents.orient_2o_chains
2′-OMe and MOE torsions scanned out of contacts (2.9 Å; 2.0 Å with H);
each chain sees the chains already turned
Offline checks
pytest (with CPPTRAJ nastruct);
smoke_tleap.py: every residue
through tleap/sander; e2e.py
8 Orientation and final checks
build._orient_and_centre · _check_result
flipped hybrids turned by Rx(180°), then centred;
O3′–P within 1.45–1.75 Å; inter-residue contacts < 2.2 Å
per record
optional
PDB file
pdbout.structure_pdb
all atoms; REMARK 999 header:
input, model, residues, data
Amber/modXNA package
amberio.package
leap.in, libs/, frcmod.modxna,
run.sh, manifest (reproduce)
High-throughput zip
batch.batch_files
pdb/<name>.pdb, report.tsv,
input.txt, README.txt